Entropy–Information Scaling of the Critical Coherence Index in Nonlinear Field Dynamics

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Abstract

Many nonlinear field systems exhibit a rich spectrum of dynamical regimes ranging from coherent structures to chaotic behaviour. In previous work the *Critical Coherence Index* (CCI) was introduced as a structural instability indicator derived from diagnostics such as dynamical activity, conservation integrity, and structural correlations [4].

In this work we investigate the physical interpretation of the CCI using numerical simulations of a one-dimensional ϕ^4 field model. We test the hypothesis that the CCI is related to the ratio between a coarse-grained entropy turnover proxy and retained structural information.

Across large ensembles of random initial conditions the CCI is consistent with a scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \propto \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}},$$

where $\dot{S}_{\mathrm{cg}}^{(+)}$ is a coarse-grained histogram activity proxy (not a thermodynamic entropy production rate; the system is Hamiltonian and true entropy production is zero by the Liouville theorem) and I_{nn} is nearest-neighbour mutual information. The best empirical scaling exponent obtained from ensemble statistics is $\alpha \approx 2.5$, yielding a Spearman correlation of $r_s = 0.734$ ($p \approx 1.6 \times 10^{-21}$). This exponent is obtained empirically from ensemble statistics and is not claimed as a fundamental constant.

These results suggest that the CCI measures the balance between structural reorganisation and retained structural information in nonlinear field dynamics.

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1 Introduction

Nonlinear field theories admit a wide range of dynamical behaviours including coherent solitonic structures, metastable patterns, and chaotic states. While the fundamental equations determine which configurations are mathematically possible, they do not directly determine which configurations are dynamically realised or persist over long timescales.

In previous work [4] we introduced the *Critical Coherence Index* (CCI) as a structural diagnostic designed to distinguish coherent from chaotic field configurations. The CCI combines measures of dynamical activity, residual instability, and structural correlations into a dimensionless instability ratio analogous to Reynolds-type numbers in fluid dynamics — specifically, a ratio of *instability* to *coherence*.

Several known physical indices share this structure:

Index	Ratio
Reynolds number	turbulence / viscosity
Lundquist number	plasma instability / resistivity
Péclet number	transport / diffusion
CCI (this work)	instability × activity / coherence

The open question addressed in this paper is:

> *Can the CCI be related to more fundamental quantities such as coarse-grained entropy turnover and structural information?*

To investigate this we analyse large ensembles of ϕ^4 lattice simulations, construct phase diagrams of the dynamical regimes, and test a power-law scaling relation between the CCI, histogram activity, and mutual information.

2 Model

We consider a one-dimensional ϕ^4 scalar field theory:

$$\ddot{\phi} = \partial_x^2 \phi - V'(\phi), \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2, \tag{1}$$

on a periodic lattice of $N = 128$ sites ($\Delta x = 1$), integrated with fourth-order Runge–Kutta ($\Delta t = 0.02$, $T_{\max} = 24$, seed $= 42$).

Initial conditions are random mixtures of four component types: vacuum configurations, kink-like structures, localised Gaussian excitations, and random noise. Mixing coefficients $a_i \sim \mathrm{Uniform}(0, 1)$ are drawn independently for each trajectory, producing a continuous ensemble of dynamical states.

All code is available at [7].

3 Structural Diagnostics

3.1 Critical Coherence Index

The CCI is defined as [4]:

$$\mathcal{C}_{\mathrm{CCI}} = \frac{\Gamma_{\mathrm{inst}}\,\Gamma_{\mathrm{prod}}}{\Gamma_{\mathrm{coh}} + \Gamma_{\mathrm{corr}} + \varepsilon}, \tag{2}$$

where Γ_{inst} measures residual instability, Γ_{prod} measures dynamical activity, and Γ_{corr} measures structural correlations.

The index separates three dynamical regimes:

$$\begin{aligned} \mathcal{C}_{\text{CCI}} < \tau_1 \quad &\Rightarrow \quad \text{ordered regime,} \\ \tau_1 \leq \mathcal{C}_{\text{CCI}} < \tau_2 \quad &\Rightarrow \quad \text{critical regime,} \\ \mathcal{C}_{\text{CCI}} \geq \tau_2 \quad &\Rightarrow \quad \text{chaotic regime,} \end{aligned}$$

with empirically calibrated thresholds $\tau_1 \approx 0.279$ and $\tau_2 \approx 0.355$.

3.2 Structural Information

Structural information is measured using nearest-neighbour mutual information:

$$I_{\text{nn}} = I(\phi_i : \phi_{i+1}), \tag{3}$$

quantifying spatial coherence between adjacent field sites.

3.3 Coarse-Grained Histogram Activity

We define a coarse-grained entropy from the field value histogram:

$$S_{\text{cg}} = -\sum_k p_k \ln p_k, \tag{4}$$

and use its positive rate of change as an activity proxy:

$$\dot{S}_{\text{cg}}^{(+)} = \max\left(0, \frac{dS_{\text{cg}}}{dt}\right). \tag{5}$$

We emphasise that $\dot{S}_{\text{cg}}^{(+)}$ is *not* a thermodynamic entropy production rate. The ϕ^4 system (1) is Hamiltonian; by the Liouville theorem, the true phase-space entropy is conserved and thermodynamic entropy production is exactly zero. $\dot{S}_{\text{cg}}^{(+)}$ measures only how rapidly the field value distribution rearranges — a proxy for dynamical activity, not dissipation.

4 Phase Diagrams

To map the dynamical regimes, we construct phase diagrams as a function of two control parameters: initial momentum amplitude σ_π and noise amplitude σ_{noise}.

Figures 1 and 2 show the mean CCI and dominant regime across the 9×9 parameter grid (8 ensemble runs per grid point).

The phase diagrams reveal a clear transition from ordered to chaotic behaviour as both σ_π and σ_{noise} increase, consistent with the regime classification of (2).

5 Entropy–Information Scaling

To test the physical interpretation of the CCI, we analyse the power-law scaling relation:

$$\mathcal{C}_{\text{CCI}} \sim \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha}}. \tag{6}$$

We scan $\alpha \in [0.5, 2.5]$ across an ensemble of 120 trajectories and compute the Spearman rank correlation between the CCI and the ratio (6) for each α. The best correlation is found at:

$$\alpha \approx 2.5. \tag{7}$$

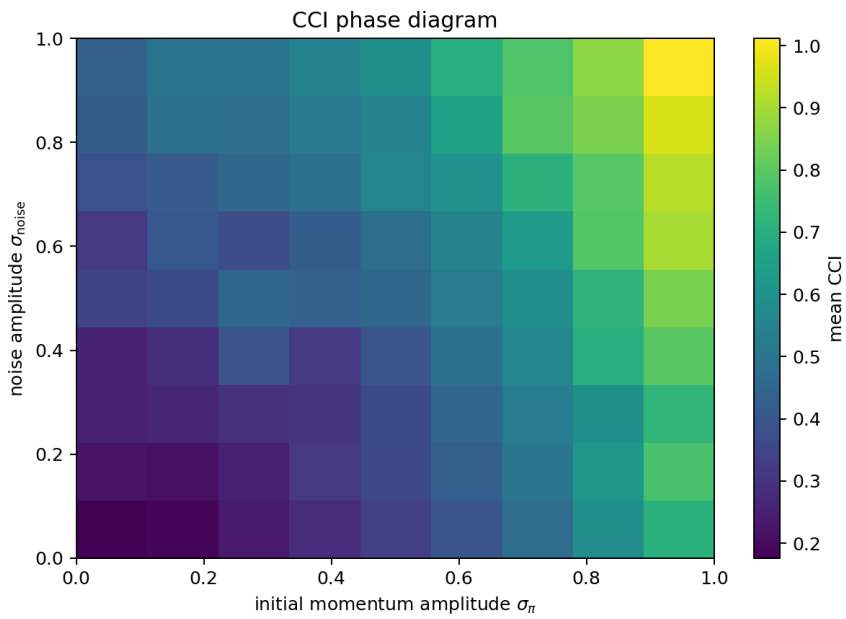

Figure 1: Mean CCI as a function of initial momentum amplitude σ_π and noise amplitude σ_{noise}. Higher values of both control parameters drive the system toward higher CCI and chaotic dynamics.

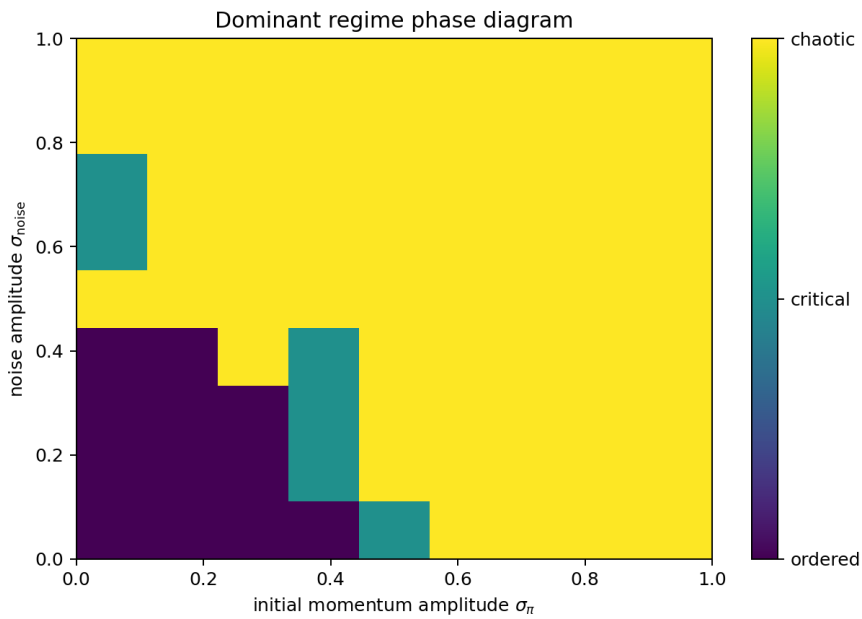

Figure 2: Dominant dynamical regime (ordered / critical / chaotic) across the $(\sigma_\pi, \sigma_{\text{noise}})$ parameter plane. A clear transition from ordered to chaotic behaviour emerges as both dynamical activity and noise increase.

We also perform a log-linear regression:

$$\ln(\mathcal{C}_{\text{CCI}}) = a_0 + b_1 \ln(\dot{S}_{\text{cg}}^{(+)}) + b_2 \ln(I_{\text{nn}}), \tag{8}$$

which provides a more direct estimate of the scaling exponents for $\dot{S}$ and I_{nn} independently.

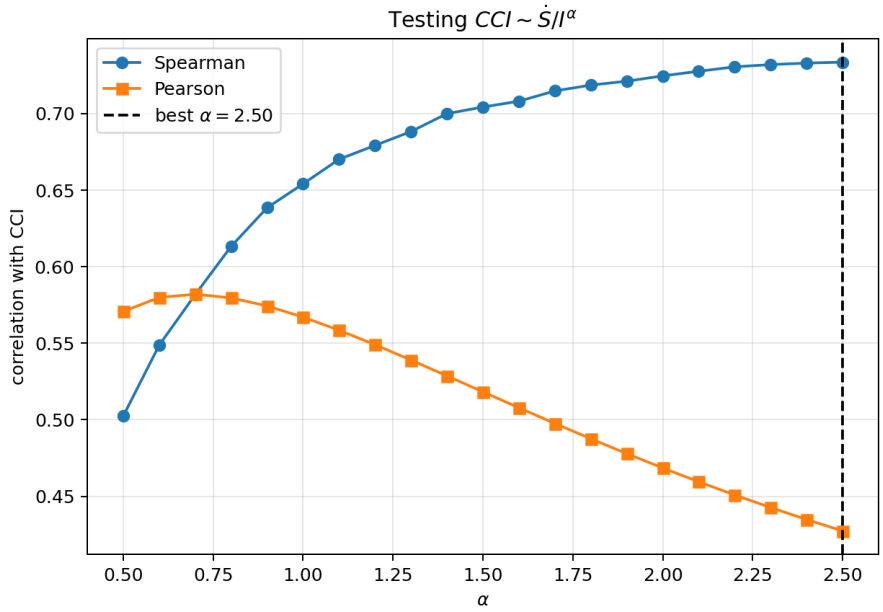

Figure 3: Spearman and Pearson correlations between the CCI and the ratio $\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ as a function of the scaling exponent α. The best correlation is obtained near $\alpha \approx 2.5$.

6 Results

The main findings are:

1. The CCI is empirically well described by the ratio $\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$. Scanning $\alpha \in [0.5, 2.5]$ reveals that the strongest correlation occurs near $\alpha \approx 2.5$, where we obtain a Spearman correlation of $r_s = 0.734$ ($p \approx 1.6 \times 10^{-21}$) and a Pearson correlation of $r = 0.43$ (Figure 3). The Spearman–Pearson discrepancy indicates a monotonic but nonlinear relationship.

2. Coarse-grained histogram activity alone does not explain the observed instability transitions; the information denominator is essential.

3. Structural mutual information I_{nn} decreases systematically as the CCI increases, consistent with the design of the index.

4. The best empirical scaling exponent is $\alpha \approx 2.5$. This exponent is obtained empirically from ensemble statistics and is *not* claimed as a theoretically derived constant. Its physical origin — whether from information scaling, dimensional analysis, or renormalisation — remains an open question.

7 Interpretation

The scaling relation (6) suggests that the CCI measures the balance between:

- **Structural reorganisation:** captured by $\dot{S}_{\mathrm{cg}}^{(+)}$, the rate at which the field value distribution rearranges.

- **Retained structural coherence:** captured by I_{nn}, the spatial mutual information between adjacent sites.

Chaos emerges when structural reorganisation grows faster than the system can maintain coherent spatial correlations. The nonlinear dependence on I_{nn} (exponent $\alpha > 1$) suggests that

structural information plays a disproportionately strong stabilising role: small reductions in I_{nn} can significantly increase the effective CCI.

This is consistent with the broader structural selection programme [1, 2, 3]: configurations with high structural information (kinks, domain walls) are robustly preferred precisely because they maintain spatial coherence against dynamical perturbations.

8 Conclusion

We have analysed the physical interpretation of the Critical Coherence Index using large ensembles of ϕ^4 field simulations, including phase diagrams over a two-dimensional control parameter space and an empirical scaling analysis.

The results indicate that the CCI is empirically well described by an entropy–information scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \propto \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \qquad \alpha \approx 2.5 \quad (r_s = 0.734,\ p \approx 1.6 \times 10^{-21}), \tag{9}$$

where the exponent is empirical and its theoretical origin remains an open problem.

This supports the interpretation of the CCI as a general structural instability parameter for nonlinear field dynamics, connecting the structural selection programme [1, 4, 5] to information-theoretic and non-equilibrium concepts.

Future work will extend the analysis to higher-dimensional systems, alternative field models (sine-Gordon, Kuramoto–Sivashinsky), and a theoretical derivation of the scaling exponent α.

A Reproducibility

All results are reproducible from the companion code:

https://github.com/Chris4081/structural-selection-principle

Key scripts:

- `cci_alpha_scaling.py` — ensemble generation, α-scan, log-fit, CCI vs ratio scatter plots.
- `cci_phase_diagram.py` — 9×9 phase diagram over $(\sigma_\pi, \sigma_{\mathrm{noise}})$.

Run:

```
pip install numpy pandas matplotlib scipy scikit-learn
python cci_alpha_scaling.py
python cci_phase_diagram.py
```

References

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